

illuminati - 2019

Advanced Chemistry Assignment-3B | Inorganic Chemistry

SINGLE CHOICE QUESTIONS

- During refining of pig iron by bessemerization process, which of the following impurity is oxidized last.
(A) Si (B) Mn (C) S (D) C
- $\text{PCl}_5 + \text{H}_2(\text{g}) \xrightarrow{\text{warm}} \text{X} + \text{Y}$
If molecular weight of 'X' is greater than that of 'Y' then compound 'X' reacts with :
(A) $\text{H}_2\text{O}(\ell)$ (B) Cl_2 gas (C) O_2 gas (D) All
- Incorrect statement for borax is :
(A) Its aqueous solution acts as buffer
(B) Its aqueous solution is used to determine strength of dil. HCl using phenolphthalein
(C) It is used as acidic flux in brazing
(D) H-bonding is present in borax
- H_2O_2 acts as oxidizing agent with which of the following reagent :
(A) KMnO_4 in basic medium (B) $\text{Fe}_2(\text{SO}_4)_3$
(C) I_2 in alkaline medium (D) HClO_4
- Milk of lime + $\text{SOCl}_2 \longrightarrow \text{P} \downarrow + \text{Q}$
Incorrect statement for compound 'P' and 'Q' is :
(A) 'Q' produces observable change with hypo
(B) 'P' dissolves in excess of H_2SO_3
(C) 'Q' produces precipitate with aqueous solution of KF
(D) 'P' reacts with H_2S gas in the presence of dilHCl
- $\text{K}_4[\text{Fe}(\text{CN})_6] + \text{conc. H}_2\text{SO}_4 \xrightarrow{\text{warm}} \text{products}$
Incorrect statement for products is :
(A) Gaseous product is used in Mond's process
(B) Oxidation state of iron remains same
(C) One of the cations is derived from nonmetals
(D) Gaseous product obtained is acidic
- $3\text{HgO} \cdot \text{HgCO}_3(\text{s}) + 2\text{PbS}(\text{s}) \xrightarrow{\Delta} \text{Products}$
Incorrect statements about products is :
(A) One of the liberated gas can decolourize aqueous KMnO_4 solution.
(B) Mercury is obtained in elemental state
(C) Lead is obtained in the form of PbO_2
(D) PbS is roasted into PbO and SO_2
- Incorrect statement for complex ions $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{MnBr}_4]^{2-}$ is :
(A) Intensity of colour in $[\text{MnBr}_4]^{2-}$ is less than $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$
(B) Both have same magnetic moment
(C) Stability : $[\text{Mn}(\text{H}_2\text{O})_6]^{2+} > [\text{MnBr}_4]^{2-}$
(D) Valence shell d- orbitals of Mn^{2+} are used in its hybridization in $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$

9. When metal cation M^{x+} forms homoleptic octahedral complex with following ligands, then for which ligand there is maximum increase in entropy.
 (A) NH_3 (B) en (C) $EDTA^{4-}$ (D) ox
10. $Na(s) + H_2(g) \xrightarrow{\Delta} P$
 $P + BF_3 \xrightarrow{180^\circ C} Q \uparrow + NaF$
 Incorrect statement for compound 'Q' is :
 (A) Its thermal decomposition liberates H_2 gas
 (B) It is an electron deficient compound
 (C) Bond order of CO decreases when it reacts with 'Q'
 (D) Four hydrogen atoms can lie in a plane of molecule 'Q'
11. In which oxyacid of phosphorus, both 'P – P' as well as 'P – H' bonds are present.
 (A) $H_4P_2O_6$ (B) $H_4P_4O_{12}$ (C) $H_4P_2O_5$ (D) None of these
12. Hydrolysis of which of the following compound does not produce gaseous product.
 (A) Ca_3P_2 (B) CaH_2 (C) Mg_3N_2 (D) CaO
13. H_2O_2 does not act as reductant with :
 (A) Acidified $KMnO_4$ solution (B) Ag_2O
 (C) H_2S gas (D) O_3
14. Nitric oxide gas is evolved by :
 (A) $Zn + dilHNO_3$ (B) $Mg + 2\% HNO_3$
 (C) $Pb + Hot dilHNO_3$ (D) $Hg + very dilute HNO_3$
15. XeF_4 does not act as fluoride donor with :
 (A) BF_3 (B) SF_6 (C) PF_5 (D) ClF_5
16. $H_3BO_3 + \text{Compound 'X'} \xrightarrow{\Delta} Y$
 If product 'Y' produces green flame, then compound 'X' cannot be :
 (A) HF (B) CH_3OH in presence of conc. H_2SO_4
 (C) $CoSO_4$ (D) All
17. Which of the following combination of reagents does not produce black precipitate :
 (A) $CuSO_4$ soln. + $PH_3(g)$ (B) $AgNO_3$ + warm soln. of $Na_2S_2O_3$
 (C) $BiCl_3 + KI$ (D) $CdCl_2 + KI$ soln.
18. Homoleptic carbonyl compound is not formed by :
 (A) Cr (B) Fe (C) V (D) Zn
19. Select correct match :

Column I		Column II	
(P)	Geometrical isomerism	1.	$[Cr(ox)(H_2O)_2(NH_3)_2]Br$
(Q)	Hydrate isomerism	2.	$[Pt(ox)(H_2O)_4]Cl_2$
(R)	Optical isomerism	3.	$[Co(NO_2)(SCN)(H_2O)_4]Br$
(S)	Ionisation isomerism	4.	$[IrBr(H_2O)(NH_3)_4]Br$

Codes :

	P	Q	R	S		P	Q	R	S
(A)	4	2	1	3	(B)	3	4	1	2
(C)	4	3	1	2	(D)	3	4	2	1

20. Select correct match :

Column I		Column II	
(P)	Sn	1.	Cupellation
(Q)	Ag	2.	Liquation
(R)	Zn	3.	Parke's Process
(S)	Pb	4.	Distillation

Codes :

	P	Q	R	S		P	Q	R	S
(A)	2	1	4	3	(B)	2	3	1	4
(C)	2	1	3	4	(D)	4	3	1	2

21. Which of the following combination of reactants does not produce hydrogen gas :
- (A) $\text{Sn} + \text{NaOH} \xrightarrow{\text{Warm}}$ (B) $\text{NaH} + \text{H}_2\text{O} \xrightarrow{\text{R.T.}}$
- (C) $\text{PH}_3 + \text{O}_2 \xrightarrow[\text{of P}_4\text{H}_4]{\text{In the presence}}$ (D) $\text{K} + \text{NH}_3(\text{liquid}) \xrightarrow{\text{Warm}}$
22. HOMO (Highest occupied M.O.) of which of the following diamagnetic molecular species is having π -bonding character.
- (A) C_2^+ (B) N_2^{2-} (C) O_2^{2-} (D) B_2^{2-}
23. Correct order C – O bond order :
- (A) $\text{CO}^+ > \text{C}_2\text{O}_4^{2-} > \text{CO}_2$ (B) $\text{CO}^+ > \text{CO}_2 > \text{C}_2\text{O}_4^{2-}$
- (C) $\text{CO}_2 > \text{CO}^+ > \text{C}_2\text{O}_4^{2-}$ (D) $\text{C}_2\text{O}_4^{2-} > \text{CO}_2 > \text{CO}^+$
24. The set representing the correct order of ionic radius is :
- (A) $\text{Na}^+ > \text{Mg}^+ > \text{Al}^{3+} > \text{Li}^+ > \text{Be}^{2+}$ (B) $\text{Na}^+ > \text{Li}^+ > \text{Mg}^{2+} > \text{Al}^{3+} > \text{Be}^{2+}$
- (C) $\text{Na}^+ > \text{Mg}^{2+} > \text{Li}^+ > \text{Al}^{3+} > \text{Be}^{2+}$ (D) $\text{Na}^+ > \text{Mg}^{2+} > \text{Li}^+ > \text{Be}^+$
25. The electronic configuration of four elements are
- (I) $[\text{Kr}]5s^1$ (II) $[\text{Rn}]5f^{14}6d^17s^2$ (III) $[\text{Ar}]3d^{10}4s^24p^5$ (IV) $[\text{Ar}]3d^64s^2$
- Consider the following statements :
- (i) I shows variable oxidation state
- (ii) II is a d-block element
- (iii) The compound formed between I and III is covalent
- (iv) IV shows single oxidation state
- Which statements is True (T) or false (F)
- (A) FTFF (B) FTFT (C) FFTF (D) FFFF
26. Which of the following statements is correct regarding following process.
- (i) $\text{Cl} \xrightarrow{\text{E.A.}} \text{Cl}^-$ (ii) $\text{Cl}^- \xrightarrow{\text{I.E.}} \text{Cl}$ (iii) $\text{Cl} \xrightarrow{\text{I.E.}} \text{Cl}^+$ (iv) $\text{Cl}^+ \xrightarrow{\text{I.E.}} \text{Cl}^{2+}$
- (A) | I.E. of process (ii) | = | E.A. of process (i) |
- (B) | I.E. of process (iii) | = | I.E. of process (ii) |
- (C) | I.E. of process (iv) | = | E.A. of process (i) |
- (D) | I.E. of process (iv) | = | I.E. of process (iii) |
27. Which of the following compounds does not have four equal bond lengths.
- (A) ClF_5 (B) XeOF_4 (C) B_2Cl_4 (D) Be_2Cl_4
28. Number of geometrical isomers of complex ion $[\text{M}(\text{AA})_2\text{b}_2\text{cd}]^{n\pm}$ in which both 'b' monodentate ligands are present at anti (trans) position of geometry.
- (A) 1 (B) 2 (C) 3 (D) None of these

29. Which charge on CO molecule would give its bond order > 3.0 ?
 (A) +1 (B) +2 (C) -1 (D) Zero
30. Increasing order of strength of hydrogen bond in the following mentioned linkages is :
 (i) $O - H \cdots S$ (ii) $S - H \cdots O$ (iii) $F - H \cdots F^-$ (iv) $F - H \cdots O$
 (A) (i) $<$ (ii) $<$ (iv) $<$ (iii) (B) (ii) $<$ (i) $<$ (iv) $<$ (iii)
 (C) (i) $<$ (ii) $<$ (iii) $<$ (iv) (D) (ii) $<$ (i) $<$ (iii) $<$ (iv)
31. Which of the following low spin complex has its magnetic moment = 2.8 B.M.
 (A) $[V(CN)_6]^{3-}$ (B) $[Ni(CN)_5]^{3-}$ (C) $[Cr(NH_3)_6]^{2+}$ (D) $[Ni(en)_3]^{2+}$
32. Which of the following is not flexidentate ligand ?
 (A) CO_3^{2-} (B) SCN^- (C) $S_2O_3^{2-}$ (D) NO_2^-
33. Consider the molecular formula of following complexes.
 (i) $CoCl_3.6NH_3$ (ii) $CoCl_3.5NH_3$ (iii) $CoCl_3.4NH_3$
 If number of coordinated NH_3 molecules in (i), (ii) and (iii) complexes respectively are 6, 5 and 4, then primary valency of metal cation in (i), (ii) and (iii) respectively are :
 (A) 6, 5, 4 (B) 3, 2, 1 (C) 0, 1, 2 (D) 3, 3, 3
34. Select the INCORRECT statements :
 (A) $B(CH_3)_3$ is a σ bonded organometallic compound
 (B) $K[PtCl_3(\eta^2 - C_2H_4)]$ is π bonded organometallic compound
 (C) Vitamin B_{12} is a coordination compound of cobalt.
 (D) $RhCl(PPh_3)_3$ (Named as Wilkinson catalyst) has sp^3 hybridised central metal cation and is used for Hydrogenation of alkenes.
35. On strongly heating, mixture of Cu_2O and Cu_2S gives :
 (A) $Cu + SO_2$ (B) $Cu + SO_3$ (C) $CuO + CuS$ (D) Cu_2SO_3
36. Polyphosphate are used as water softening agents because they :
 (A) form soluble complexes with anionic species
 (B) precipitate anionic species
 (C) form soluble complexes with cationic species
 (D) precipitate cationic species
37. Reduction of $BaSO_4$ with carbon gives :
 (A) $BaSO_3$ (B) BaS (C) BaS_2O_3 (D) Ba
38. $MgBr_2$ and MgI_2 are soluble in acetone because of :
 (A) Their ionic nature (B) Their more covalent nature
 (C) Their metallic nature (D) None of these
39. Which of the following is not responsible for colour of the compound
 (A) Polarization (B) d-d transition
 (C) Charge transfer spectra (D) Dipole moment
40. In which of the following molecule central atom has higher % s-character in its bond pair ?
 (A) AsH_3 (B) GeH_4 (C) P_4 (D) H_2Se

MULTIPLE CHOICE QUESTIONS

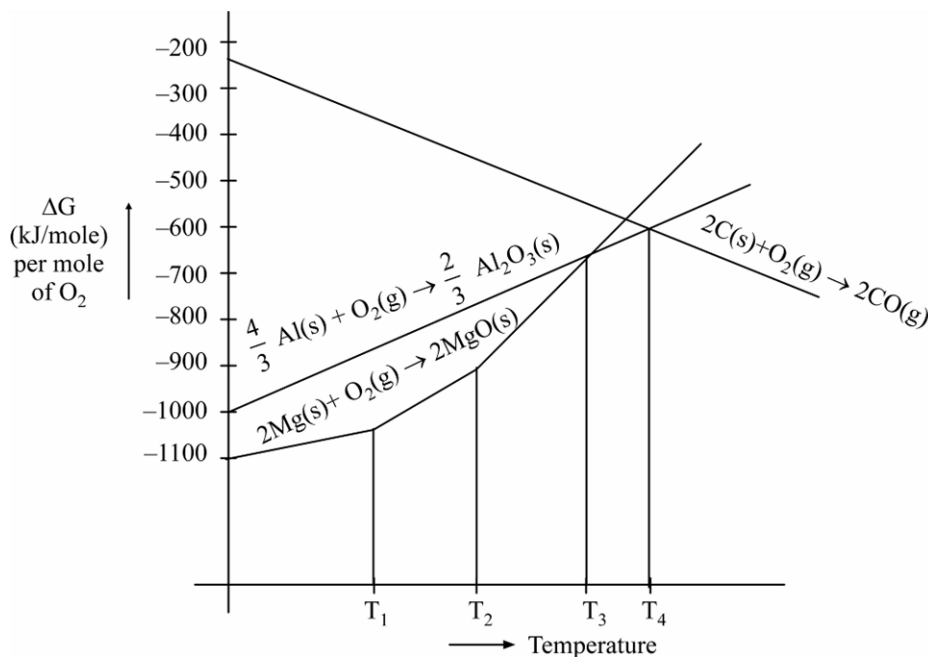
41. KO_2 produces observable change with :
 (A) $\text{Mg}^{2+}(\text{aq})$ (B) $\text{Fe}^{2+}(\text{aq})$ (C) $\text{Ba}^{2+}(\text{aq})$ (D) $\text{Cr}^{3+}(\text{aq})$
42. Lowest Unoccupied Molecular Orbital (LUMO) of which of the following molecular species has centre of symmetry :
 (A) N_2 (B) C_2 (C) O_2 (D) Li_2
43. Correct statement(s) for complex compound $\text{CoBrCl}_2 \cdot 2\text{en}$ is/are :
 (A) It exhibits ionization isomerization
 (B) One of the geometrical isomers exhibits optical activity
 (C) All stereoisomers exhibit optical activity
 (D) Its structural isomers can be distinguished by AgNO_3 solution
44. $\text{Fe}^{3+}(\text{aq}) + \text{Sn}^{2+}(\text{aq}) \longrightarrow \text{P} + \text{Q}$
 If aqueous solution of cation 'Q' reacts with H_2S gas, then which of the following reagent(s) can be used to identify cation 'P' from its aqueous solution :
 (A) dil. HNO_3 (B) $\text{K}_3[\text{Fe}(\text{CN})_6]$ (C) Na_2O_2 (D) SO_2 gas
45. Complexes $[\text{Ni}(\text{dmg})_2]$ and $[\text{Ni}(\text{NH}_3)_6]^{2+}$ differ in following characteristics :
 (A) Colour
 (B) Stability
 (C) Magnetic moment
 (D) Maximum number of N-atoms that lie in a plane is equal to four
46. Which of the following process is/are common in the commercial extraction of lead and copper :
 (A) Self reduction (B) Poling
 (C) Electrolytic refining (D) Parke's process
47. For reactions :
 $\text{SO}_2(\text{g}) + \text{Cl}_2(\text{g}) \xrightarrow{\text{R.T.}} \text{X}$
 $\text{X} + \text{P}_4 \xrightarrow{\text{warm}} \text{Y} + \text{Z}$
 $\text{Y} + \text{P}_4\text{O}_{10} \longrightarrow \text{POCl}_3$
 Compounds 'X', 'Y' and 'Z' are :
 (A) SO_2Cl_2 (B) PCl_3 (C) SO_2 (D) SOCl_2
48. Pair(s) of co-ordination compounds that have different colour :
 (A) $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$, $[\text{Ni}(\text{NH}_3)_6]^{2+}$ (B) $[\text{NiCl}_4]^{2-}$, $[\text{CuCl}_4]^{3-}$
 (C) $[\text{Zn}(\text{NH}_3)_4]^{2+}$, $[\text{Cd}(\text{CN})_4]^{2-}$ (D) Cis - $[\text{CoCl}_2(\text{NH}_3)_4]\text{Cl}$, trans - $[\text{CoCl}_2(\text{NH}_3)_4]\text{Cl}$
49. Statement(s) which favour zinc metal to be used as solvent in Parke's process.
 (A) For any composition, zinc and lead do not attain a common melting point.
 (B) Zinc can be recovered from zinc-silver layer by distillation
 (C) Zinc is less denser than lead
 (D) Silver is more soluble in lead than in zinc.
50. Which of the following combination(s) of reagents can produce halogen/pseudohalogen ?
 (A) $\text{CaBr}_2 + \text{conc. H}_2\text{SO}_4$ (B) $\text{Fe}^{3+}(\text{aq}) + \text{KI}$
 (C) $\text{Cu}^{2+}(\text{aq}) + \text{KCN}$ (D) $\text{CrO}_3 + \text{dilHBr}$
51. Correct statement(s) for complex compound $[\text{CoBr}(\text{NCS})(\text{H}_2\text{O})(\text{NH}_3)_3](\text{NO}_2)$
 (A) Its Ionization isomer is diamagnetic
 (B) It has one pair of enantiomers
 (C) In one of its optical isomers, Br^- and NCS^- are at trans position
 (D) It is less stable than $[\text{Co}(\text{en})_3]^{3+}$.

52. Which of the following stage(s) are involved in the extraction of blister copper from chalcopyrite :
- (A) Bessemerization (B) Roasting
(C) Poling (D) Slag formation
53. Which of the following compound(s) on thermal decomposition produce residue that does not oxidize HCl acid ?
- (A) $K_2SO_4 \cdot Cr_2(SO_4)_3 \cdot 24H_2O$ (B) $KMnO_4$
(C) $FeSO_4 \cdot 7H_2O$ (D) $2CuCO_3 \cdot Cu(OH)_2$
54. Acidic character of HF increases appreciably on mixing it with :
- (A) XeF_2 (B) BF_3 (C) PF_5 (D) SiF_4
55. Residue of metallic silver is obtained when, which of the following compounds is added into aqueous $AgNO_3$ solution :
- (A) H_3PO_2 (B) Adding $Na_2S_2O_3$ followed by warming
(C) Adding Na_2SO_3 followed by boiling (D) Addition Na_2CO_3 followed by boiling
56. $PCl_5 + Cu \xrightarrow{\Delta} X + Y$
If aqueous solution of Y is acidic then correct statement(s) is/are :
- (A) one of the hydrolysed product of 'Y' undergoes tautomerism
(B) On passing NH_3 gas into solution of 'X', its colour is intensified
(C) 'X' gives chocolate brown colour ppt with $K_4[Fe(CN)_6]$
(D) 'X' produces precipitates with NH_4SCN
57. Blue colouration is observed in presence of starch for which of the following reaction(s) :
- (A) $I^-(aq) + H^+(aq) \xrightarrow{\text{atm air}}$ (B) $I^-(aq) + Fe^{3+}(aq) \longrightarrow$
(C) $IO_3^-(aq) + SO_2(g) \xrightarrow{\text{limited}}$ (D) $I^-(aq) + NO_2^-(aq) + H^+(aq) \longrightarrow$
58. Which of the following compound(s) can give precipitate with solution of chrome alum :
- (A) $BaCl_2$ (B) Na_2CO_3 (C) $(NH_4)_2S$ (D) excess KOH
59. For reaction : $Sn + 2PCl_5 \xrightarrow{\Delta} X + 2Y$
Correct statement(s) is/are :
- (A) $HgCl_2$ changes to black residue with excess of X
(B) Oxyacid obtained from hydrolysis of Y, can undergo tautomeric change
(C) 'X' is soluble in excess of NaOH
(D) Y is polar and has plane of symmetry
60. Which of the following structure is planar ?
- (A) $Na_3B_3O_6$ (B) I_2Cl_6
(C) Sheet silicate (D) Inorganic graphite layer
61. Pair(s) of cations that can be separated by NH_4OH solution.
- (A) $Pb^{2+}(aq), Cu^{2+}(aq)$ (B) $Hg^{2+}(aq), Ag^+(aq)$
(C) $Fe^{3+}(aq), Mg^{2+}(aq)$ (D) $Ni^{2+}(aq), Mn^{2+}(aq)$
62. $M^{n+}(aq) \xrightarrow{KI} \underset{\text{(coloured ppt)}}{X \downarrow} \xrightarrow[\text{(colourless solution)}]{\text{excess KI}} Y$
Which of the following cation(s) respond to the above reaction ?
- (A) $Ag^+(aq)$ (B) $Pb^{2+}(aq)$ (C) $Bi^{3+}(aq)$ (D) $Hg^{2+}(aq)$
63. Salt (solid) + conc. $H_2SO_4 \xrightarrow{\text{warm}} \underset{\text{(Coloured gas)}}{X \uparrow}$
If gas 'X' undergoes disproportionation in NaOH solution, then which of the following salt(s) can respond to the above reaction :
- (A) NaCl (B) $NaNO_2$ (C) $NaNO_3$ (D) NaBr

64. Select the correct order of periodic properties of species :
- (A) $\text{Fe}^{2+} < \text{Fe}^{3+}$: ionic radii (B) $\text{N} < \text{O}$: second ionization energy
(C) $\text{Cu} < \text{Zn}$: Atomic Radius (D) $\text{In} < \text{Tl}$: first ionization energy
65. Correct match is/are :
- (A) $\text{AlN} > \text{MgO} > \text{MgF}_2$: Lattice energy
(B) $\text{N}_2 > \text{N} > \text{O} > \text{O}_2$: Ionisation energy
(C) $\text{SF}_6 > \text{PF}_5 > \text{SiF}_4$: Lewis acidic character
(D) $\text{SiCl}_4 > \text{SiBr}_4 > \text{SiI}_4$: Electronegativity of Si
66. Incorrect order of bond angle is :
- (A) $\text{OCl}_2 > \text{SF}_2 > \text{AsH}_3 > \text{H}_2\text{Se}$ (B) $\text{NH}_3 > \text{PF}_3 > \text{PH}_3 > \text{H}_2\text{S}$
(C) $\text{XeO}_4 > \text{ClO}_4^- > \text{SO}_4^{2-} > \text{CF}_4$ (D) $\text{N}(\text{SiH}_3)_3 > \text{NH}_3 > \text{NMe}_3$
67. Which of the following order is/are Correct for the given properties indicated against each ?
- (A) $[\text{Cr}(\text{NH}_3)_6]^{3+} > [\text{Mn}(\text{CN})_6]^{3-} > [\text{V}(\text{CO})_6]$: Magnetic moment (spin only value)
(B) $[\text{Co}(\text{CN})_6]^{3-} > [\text{Co}(\text{NH}_3)_6]^{3+} > [\text{Co}(\text{H}_2\text{O})_6]^{3+}$: Δ_0 values
(C) $[\text{Ni}(\text{CO})_4] > [\text{Co}(\text{CO})_4]^- > [\text{Fe}(\text{CO})_4]^{2-}$: CO bond order
(D) $[\text{Fe}(\text{CN})_6]^{3-} > [\text{Co}(\text{CN})_6]^{4-} > [\text{Ni}(\text{NH}_3)_6]^{2+}$: Effective atomic number of metal ion
68. For reaction : $\text{NaF} + \text{XeF}_4 \longrightarrow \text{P}$, the correct statement(s) for anionic part of compound 'P' is/are :
- (A) Six atoms lie in a plane
(B) Non polar
(C) Non axial d-orbital used in hybridization of xenon has two xz and yz nodal plnes.
(D) All bonds are equivalent
69. Octahedral complex which neither show ionization nor hydrate isomerism :
- (A) $[\text{Fe}(\text{H}_2\text{O})_5\text{NH}_3]\text{SO}_4$ (B) $[\text{Fe}(\text{H}_2\text{O})_5\text{Cl}]\text{SO}_4$
(C) $[\text{Co}(\text{NH}_3)_4\text{SO}_4]\text{Cl}$ (D) $[\text{Fe}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}$
70. Which of the following set of species correctly represent sp^3 hybridization of central atom and tetrahedral geometry ?
- (A) ClO_4^- , SO_4^{2-} , Diamond, $[\text{Zn}(\text{H}_2\text{O})_4]^{2+}$ (B) SiO_4^{4-} , $[\text{PCl}_4]^+$, SiC, HClO_4
(C) $\text{Ni}(\text{CO})_4$, SnCl_4 , Graphite, Ice (D) P_4 , $[\text{PtCl}_4]^{2-}$, SiF_4 , $[\text{Be}(\text{gly})_2]^{2+}$
71. Consider the following steps : $\text{Cu}_2\text{S} \xrightarrow{\text{roast in air}} \text{products (X)} \xrightarrow{\text{heating without air}} \text{Products (Y)}$
Select Correct statement(s) :
- (A) Self – reduction is involved in above step.
(B) Disproportionation reaction ($\text{Cu}_2\text{S} \xrightarrow{\Delta} \text{Cu} + \text{CuS}$) is involved in above step.
(C) Products (X) is a mixture of Cu_2O and Cu_2S
(D) Products (Y) is a mixture of Cu and SO_2
72. The complex $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]^{2+}$ is formed in the brown ring test for nitrates when freshly prepared FeSO_4 solution is added to aqueous solution of NO_3^- followed by addition of conc. H_2SO_4 . Select the correct statement about this complex :
- (A) Colour observed is due to charge transfer
(B) It has iron in +1 oxidation state and nitrosyl as NO^+
(C) It has magnetic moment of 3.87 BM confirming three unpaired electrons in Fe
(D) CFSE value is equal to $-0.8\Delta_0$
73. Reaction involves an increases in bond length :
- (A) $\text{O}_2 \longrightarrow \text{O}_2^+$ (B) $\text{O}_2 \longrightarrow \text{O}_2^-$ (C) $\text{N}_2 \longrightarrow \text{N}_2^+$ (D) $\text{B}_2 \longrightarrow \text{B}_2^{2-}$

74. CrO_2Cl_2 (vapour) $\xrightarrow{\text{excess NaOH solution}}$ X (Coloured soln.) $\xrightarrow{\text{Y Reagent}}$ Z $\downarrow$ (Coloured ppt)
 then reagent (Y) can be :
 (A) $\text{Sr}(\text{NO}_3)_2$ (B) $\text{Pb}(\text{NO}_3)_2$ (C) AgNO_3 (D) $\text{Ba}(\text{NO}_3)_2$

75.



According to the given Ellingham diagram the correct option(s) is/are :

- (A) Above T_3 , reaction $3\text{MgO} + 2\text{Al} \xrightarrow{\Delta} \text{Al}_2\text{O}_3 + 3\text{Mg}$ occurs
 (B) Carbon reduction of MgO is faster than that of Al_2O_3 above T_4 temperature
 (C) Above T_4 , Mg metal is extracted in vapour phase by carbon reduction of MgO
 (D) Carbides of Al and Mg formed above T_4 . On hydrolysis do not liberate C_2H_2 .
76. Which of the following complex(es) has / have only two stereoisomers :
 (A) $[\text{PdBrCl}(\text{NO}_2)\text{NH}_3]^-$ (B) $[\text{Zn}(\text{SCN})(\text{H}_2\text{O})(\text{Gly})]$
 (C) $[\text{Co}(\text{acac})_3]$ (D) $[\text{CrBr}_3(\text{H}_2\text{O})(\text{en})]$
77. In which of the following complex(es) unpaired electron is lying in d-orbital of valance shell of metal cation ?
 (A) $[\text{Cu}(\text{en})_2]^{2+}$ (B) $\text{Fe}(\text{CO})_5$ (C) $[\text{Co}(\text{CN})_6]^{4-}$ (D) $[\text{Cu}(\text{NO}_2)_5]^{3-}$
78. During the production of steel from Haematite which of the following statements is/are correct.
 (A) Fe_2O_3 is reduced to Fe by carbon in reduction zone of blast furnance.
 (B) Impurities of impure iron are removed by oxidation using Bessemer's converter.
 (C) spiegeleisan is added to wrought iron
 (D) Phosphorus impurities are converted into slag which is used as fertilizer
79. In which of the following reaction(s) H_2O_2 acts reducing agent ?
 (A) $\text{H}_2\text{O}_2 + \text{Cr}_2\text{O}_7^{2-}(\text{aq}) + \text{H}^+(\text{aq}) \longrightarrow \text{Cr}^{3+}(\text{aq}) + \text{O}_2 \uparrow + \text{H}_2\text{O}$
 (B) $\text{H}_2\text{O}_2 + \text{Cr}^{3+}(\text{aq}) + \text{OH}^-(\text{aq}) \longrightarrow \text{CrO}_4^{2-}(\text{aq})$
 (C) $\text{Ag}_2\text{O} + \text{H}_2\text{O}_2 \longrightarrow 2\text{Ag} + \text{O}_2 \uparrow + \text{H}_2\text{O}$
 (D) $\text{H}_2\text{O}_2 + \text{MnO}_4^-(\text{aq}) + \text{H}^+ \longrightarrow \text{Mn}^{2+}(\text{aq}) + \text{O}_2 \uparrow$
80. $[(\text{H}_2\text{O})_4\text{Co}(\text{O}_2)_2\text{Co}(\text{H}_2\text{O})_4](\text{SO}_4)_2 \xrightarrow{[\text{x}] \text{ reagent}} [(\text{H}_2\text{O})_4\text{Co}(\text{O}_2)_2\text{Co}(\text{H}_2\text{O})_4](\text{SO}_4)$
 (I) (II)
 In both the complex, Co has $t_{2g}^6 e_g^0$ configuration. Which option is CORRECT ?
 (A) Complex (I) is Paramagnetic (B) Complex (II) is Diamagnetic
 (C) (X) is oxidizing agent (D) (X) is reducing agent

MATCHING MATCH TYPE

81. Match the following :

Column-I (species)	Column-II (Characteristics)
(A) XeF_5^-	(P) Non-polar and planar
(B) MnO_4^-	(Q) Species having equal bond angle and bond length
(C) $\cdot\dot{\text{C}}\text{H}_3$	(R) Both axial d-orbitals are involved in hybridisation
(D) I_2Cl_6	(S) All non-axial d-orbitals are involved in hybridisation

82.

Column-I (Compounds)	Column-II (Characteristics)
(A) PH_3	(P) Exists in gaseous state at room temperature
(B) B_2H_6	(Q) Reacts with aqueous CuSO_4 solution
(C) NH_3	(R) Molecule acts as lewis base
(D) H_2S	(S) On hydrolysis hydrogen gas is liberated.
	(T) Can exhibit reducing character.

83.

Column-I (Metal)	Column-II (Characteristics)
(A) Cu	(P) Self reduction is used for commercial extraction.
(B) Hg	(Q) CO gas can reduce metal oxide
(C) Zn	(R) Metal sulphide is soluble in excess of KCN solution.
(D) Fe	(S) Carbonate ore of metal is found
	(T) $\text{M}_{(\text{aq})}^{x+}$ ion oxidizes KI solution into I_2 .

84.

Column-I (Observations)	Column-II (Compounds)
(A) Coloured fumes are evolved on warming with conc. H_2SO_4	(P) $\text{Na}(\text{NH}_4)\text{HPO}_4 \cdot 4\text{H}_2\text{O}$
(B) Ammonia gas is evolved on heating	(Q) NH_4I
(C) Precipitate is formed with AgNO_3 solution	(R) KBr
(D) Coloured residue is obtained on heating	(S) $\text{Pb}(\text{NO}_3)_2$
	(T) $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$

85.

Column-I (complex compounds)	Column-II (Type of isomerism)
(A) $[\text{Ir}(\text{NO}_2)(\text{H}_2\text{O})_5]\text{Br}_2$	(P) Hydrate isomerism
(B) $[\text{CoBr}(\text{SCN})(\text{en})_2](\text{NO}_2)$	(Q) Cis isomers are optically active
(C) $[\text{Cr}(\text{acac})(\text{H}_2\text{O})_4](\text{NO}_3)_2$	(R) Trans isomer(s) is/are optical isomers
(D) $[\text{PtCl}(\text{ONO})(\text{H}_2\text{O})_2(\text{en})]\text{Cl}_2$	(S) Linkage isomerism
	(T) Does not exhibit geometrical isomerism

86.

Column-I [process used for commercial extraction of pure metal]	Column-II (Metals)
(A) Self reduction	(P) Fe
(B) Carbon reduction	(Q) Cu
(C) Distillation	(R) Pb
(D) Bessemerization	(S) Hg
	(T) Zn

87.

Column-I (Observations)	Column-II (Reagents)
(A) Precipitate with $\text{Pb}(\text{CH}_3\text{COO})_2$	(P) CuSO_4
(B) Formation of precipitate with KI	(Q) FeCl_3
(C) Complex formation with $\text{Na}_2\text{S}_2\text{O}_3$ solution	(R) AgNO_3
(D) Liberation of gas with $(\text{Zn} + \text{NaOH})$	(S) $\text{Hg}(\text{NO}_3)_2$
	(T) BiCl_3

88.

Column-I (Observations)	Column-II (Salt)
(A) Conc. H_2SO_4 plays its two roles	(P) NaBr
(B) Liberation of coloured gas/vapour with $\text{K}_2\text{Cr}_2\text{O}_7(\text{s}) + \text{Hot conc. H}_2\text{SO}_4$	(Q) Na_2S
(C) Acidified KMnO_4 is decolourized	(R) $\text{Na}_2\text{C}_2\text{O}_4$
(D) Vapour/gas is evolved on warming with dil H_2SO_4 and MnO_2	(S) NaCl
	(T) NaI

89.

Column I (Compound)	Column II (observations of Thermal decomposition)
(A) FeC_2O_4	(P) Coloured metal oxide is obtained as residue
(B) $\text{Mg}(\text{NO}_3)_2$	(Q) Metal cation is involved in redox reaction
(C) FeSO_4	(R) Paramagnetic gas is evolved
(D) $\text{HgCO}_3 \cdot \text{Hg}(\text{OH})_2$	(S) Metallic residue is obtained finally
	(T) Acidic gas is evolved

90.

Column-I (Same characteristic of given pair)	Column-II (Pair of complexes)
(A) Geometry of complexes	(P) $[\text{Cu}(\text{en})_2]^{2+}$ and $[\text{Ni}(\text{dmg})_2]$
(B) Type of Hybridisation	(Q) $[\text{NiF}_6]^{2-}$ and $[\text{IrF}_6]^{3-}$
(C) Low spin complex	(R) $[\text{Co}(\text{CO})_4]^-$ and $[\text{Fe}(\text{CO})_4]^{2-}$
(D) Magnetic behaviour	(S) $[\text{Ni}(\text{en})_3]^{2+}$ and $[\text{Cr}(\text{NH}_3)_6]^{2+}$
	(T) $[\text{CuCl}_5]^{3-}$ and $[\text{MnBr}_5]^{3-}$

91.

Column-I	Column-II
(A) $[\text{Co}(\text{NO}_3)(\text{OH})(\text{H}_2\text{O})_2(\text{NH}_3)_2]\text{Br}$	(P) Linkage isomerism
(B) $[\text{Pt}(\text{SCN})_2(\text{en})(\text{Py})_2]\text{C}_2\text{O}_4$	(Q) Ionization isomerism
(C) $[\text{PtBr}(\text{SCN})(\text{H}_2\text{O})(\text{NH}_3)(\text{Py})_2]\text{Br}_2$	(R) Even numbered optical isomers
(D) $[\text{PdCl}(\text{S}_2\text{O}_3)(\text{en})(\text{NH}_3)_2]\text{Cl}$	(S) Trans-isomers are greater than cis-isomers
	(T) Cis-isomers are greater than trans-isomers

92.

Column-I	Column-II
(A) Diamagnetic gas is evolved on warming with conc. H_2SO_4	(P) $\text{Br}^- (\text{aq})$
(B) Acidified solution of given ion turns (KI + starch) solution blue	(Q) $\text{S}_2\text{O}_3^{2-} (\text{aq})$
(C) Reacts with Cl_2 water	(R) $\text{NO}_2^- (\text{aq})$
(D) Ag^+ Salt of given ion dissolves in excess NH_3 solution	(S) $\text{S}^{2-} (\text{aq})$
	(T) $\text{C}_2\text{O}_4^{2-} (\text{aq})$

93.

Column-I	Column-II
(A) Chemical change is observed with excess Na_2O_2	(P) FeSO_4
(B) Metal sulphide ppt is formed with $(\text{NH}_4)_2\text{S}$ solution	(Q) $\text{Pb}(\text{NO}_3)_2$
(C) Precipitate is formed with excess NaOH	(R) MnSO_4
(D) White ppt is formed with BaCl_2 solution	(S) AgNO_3
	(T) $\text{Cr}_2(\text{SO}_4)_3$

94.

Column-I	Column-II
(A) Purification by oxidation method	(P) Fe
(B) Carbon reduction method applied for commercial extraction of metal	(Q) Zn
(C) Froth floatation method	(R) Cu
(D) Electrolytic refining method	(S) Ag
	(T) Al

95.

Column-I	Column-II
(A) Only three halogen atoms are lying in a plane	(P) IO_2F_2^-
(B) Fractional bond order is between central atom and oxygen atom	(Q) XeO_6^{4-}
(C) $p\pi - d\pi$ bond is present	(R) POCl_3
(D) d-orbital having no nodal plane is used in hybridization of central atom	(S) CrO_2F_2
	(T) SOF_4

96.

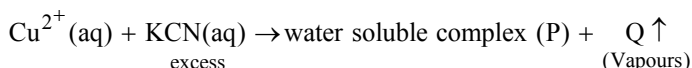
Column-I	Column-II
(A) Undergoes complete hydrolysis with excess water	(P) ClF_5
(B) Can react with CsF	(Q) BF_3
(C) Molecule having only one type of bond angle	(R) XeF_6
(D) One of the final hydrolysed product reacts with silica present in glass	(S) SF_4
	(T) SiF_4

97.

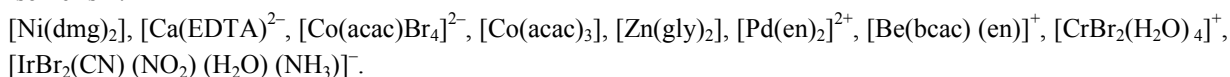
Column-I	Column-II
(A) Acid can undergo tautomeric change	(P) H_3PO_3
(B) Undergoes disproportionation on heating	(Q) HNO_2
(C) Changes colour of FeSO_4 solution	(R) H_2SO_3
(D) Decolourizes acidified KMnO_4 solution	(S) HClO_4
	(T) $\text{H}_2\text{C}_2\text{O}_4$

SUBJECTIVE TYPE

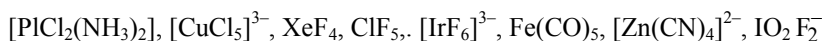
98. Find out total number of C – Cu – C bond angle(s) in water soluble complex 'P' in following reaction



99. Which of the following complex ions/compounds does/do NOT exhibit geometrical isomerism but exhibit optical isomerism.



100. In which of the following molecules or ions, dz^2 orbital is used in bond formation.



101. For reaction : $\text{XeF}_6 + \text{OH}^-(\text{aq}) \xrightarrow{\text{R.T.}} \text{Products}$

Find out the sum of oxidation state of xenon atoms in products.

102. Consider the following reaction $\text{P}_4 + \text{KOH} \longrightarrow \text{PH}_3 + \text{X}$

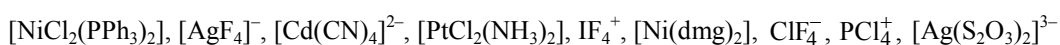
How many P – H bonds are present in species X ?

103. Find the value of expression $[X - Y]$ of IF_5 molecule.

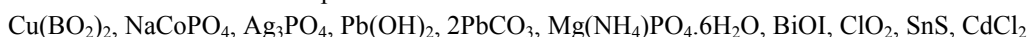
Where, X = total number of bond angle $\angle \text{F} - \text{I} - \text{F}$, that are less than 90° .

Y = maximum number of atom(s) that lie in a plane having iodine atom,

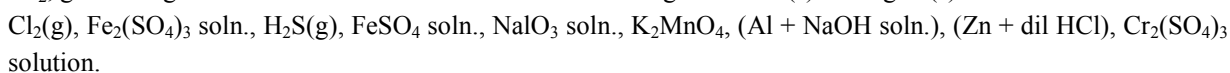
104. Find out total number of species from the following which is/are tetrahedral w.r. to their respective central atom/ion.



105. Find out total number of compounds which are coloured :



106. SO_2 , gas undergoes redox reaction with which of the following substance(s) or reagent(s)



107. Find out total number of nonmetals from following which react with water.
 $N_2, S_8, F_2, I_2, P_4, Br_2, \text{Boron}, Si, Cl_2$
108. Find out maximum number of carbon atoms that can lie in a plane in $[Cu(gly)_2]$
109. Find the number of species / molecules having at least four identical bond lengths.
 $XeO_6^{4-}, H_3\overset{+}{N}NH_2, TeCl_4, AsH_4^+, XeF_4, N_2O_4, I_2Cl_6, B_2H_6, XeF_5^+$
110. $NO_3^-(aq) + Zn + NaOH \longrightarrow [Zn(OH)_4]^{2-} + X \uparrow (\text{gas})$
 $X(\text{gas}) + Mn^{2+}(aq) + H_2O_2 \longrightarrow Y \downarrow (\text{Brown ppt})$
 Oxidation state of Mn in brown ppt is
111. From the following cations, find out total number of metal cation(s) which produce black sulphide on passing H_2S gas into their aqueous salt solution.
 $Hg^{2+}(aq), Pb^{2+}(aq), Cu^{2+}(aq), Mn^{2+}(aq), Fe^{2+}(aq), Ni^{2+}(aq), Co^{2+}(aq), Fe^{3+}(aq), Cd^{2+}(aq)$
112. Consider the complete hydrolysis of following compounds
 $\text{Marshal's acid} + xH_2O \xrightarrow{\text{complete hydrolysis}}$
 $\text{Xenon oxytetrafluoride} + yH_2O \xrightarrow{\text{complete hydrolysis}}$
 Then calculate the value of $[x + y]$
113. From the following given species, find out total number of species having partially filled HOMO (Highest Occupied Molecular Orbital) of sigma character.
 $N_2^-, N_2^+, O_2, C_2^{2-}, B_2, H_2^+, CO, NO, O_2^+$
114. Find out number of geometrical isomers of complex ion $[CrBrCl(NO_2)_2(en)]^-$, in which both NO_2^- ligands are present at cis position of geometry.
115. A tetrahedron complex ion having molecular formula $[M(CO)_n]^{x\pm}$ follows Sidwick's rule of EAN and metal 'M' belongs to first transition series and 9th group. Find out number of unpaired electron(s) present in the carbonyl is :
116. $B_2O_3 + 3H_2 + 2Al \xrightarrow[150^\circ C]{750 \text{ atm}} 'X' \xleftarrow{\text{ether}} 2BF_3 + 6NaH$
 Then maximum number of H atom(s) that can lie in same plane in compound 'X' is
117. Total number of compound(s) having 'X - X' linkage, (where X is 15th group element)
 $N_2O_4, N_2O, P_4O_6, N_2O_3, N_2H_4, P_4S_{10}, N_2O_5, H_4P_2O_6, H_2N_2O_2$.
118. How many molecular species will produce O_2 on reaction with water at room temperature.
 $CrO_5, XeOF_4, I_2, XeF_4, Br_2, XeF_2, Cl_2, Na_2O_2, F_2$.
119. When Cl_2 water is dropwise added into KI solution, initially brown colouration appears, further on adding excess Cl_2 water brown colouration disappears due to formation of colourless compound (X) predict oxidation state of central atom in compound (X).
120. Find out total number of ion(s)/compound(s) from the following, which can produce blue colour with starch iodide solution.
 $Cu^{2+}(aq), HNO_2, H_2S, Br_2, Fe^{3+}(aq), H_3PO_4, Na_2O_2, H_3BO_3, Bi^{3+}(aq)$
121. Total number of molecular species which has/have open book like structure.
 $Na_2O_2, S_2F_2, O_2F_2, S_2O_8^{2-}, P_2O_7^{4-}, H_2N_2O_2, S_2C_2N_2, C_2N_2, S_4O_6^{2-}$
122. Consider thermal decomposition of following ammonium salts and calculate value of expression $\frac{|P - Q|}{2}$
 $(NH_4)_2Cr_2O_7, NH_4NO_2, NH_4NO_3, NH_4Cl, CH_3COONH_4, NH_4ClO_4, (NH_4)_2S, (NH_4)_2CO_3, (NH_4)_2C_2O_4$
 Where : P : Total number of ammonium salts which liberates basic NH_3 gas on thermal decomposition.
 Q : Total number of ammonium salts which liberates neutral N_2 gas on thermal decomposition.

123. Consider the following reaction and predict the number of d-electrons in t_{2g} set of d-orbitals in purple complex.
 $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}] + \text{Na}_2\text{S} \longrightarrow \text{purple complex}$
124. Total number of compounds, which contains only one peroxy linkage
 $\text{PbO}_2, \text{BaO}_2, \text{H}_2\text{O}_2, \text{O}_2\text{F}_2, \text{O}_2[\text{PtF}_6], \text{H}_2\text{S}_2\text{O}_8, \text{H}_2\text{SO}_5, \text{Na}_2\text{O}_2, \text{CrO}_5$
125. $\text{NO}_2 + x\text{H}_2\text{O} \xrightarrow{\text{Hydrolysis}} \text{ClO}_3 + y\text{H}_2\text{O} \xrightarrow{\text{Hydrolysis}}$
 Then calculate the value of $[x + y]$.
126. $\text{ClO}_2 + \text{H}_2\text{O} \longrightarrow x + y$
 Calculate difference of oxidation states of chlorine atom in compounds x and y formed after hydrolysis.
127. Consider P_4O_{10} molecule and then calculate the value of expression $\left[\frac{axb}{c} \right]$
 Where
 a = Total number of P – O – P bonds
 b = Total number of σ coordinate bonds
 c = Total number of π dative bonds (back bonds)
128. The number of radicals which give coloured product on observation with conc. H_2SO_4 .
 $\text{CO}_3^{2-}, \text{NO}_2^-, \text{Cl}^-, \text{Br}^-, \text{I}^-, \text{NO}_3^-, \text{PO}_4^{3-}, \text{CrO}_4^{2-}, \text{SO}_3^{2-}$
129. Calculate number of compound which contain direct single covalent bond (x – x) between same atoms.
 $\text{P}_4\text{O}_6, \text{H}_2\text{S}_2\text{O}_4, \text{H}_2\text{S}_2\text{O}_5, \text{H}_2\text{S}_2\text{O}_6, \text{H}_2\text{O}_2, \text{H}_4\text{P}_2\text{O}_8, \text{H}_4\text{P}_2\text{O}_7, \text{H}_2\text{SO}_5, \text{H}_4\text{P}_2\text{O}_6$.
130. When black iron sulphide is roasted in air, gas P is formed, further gas P on reaction with sufficient amount of Cl_2 produces another compounds Q. Then find out difference in oxidation state between sulphur atoms in gas P and compound Q.
131. Calculate sum of unpaired electrons in central metal ions of
 $\text{Fe}(\text{CN})_6^{4-}, [\text{FeF}_6]^{3-}, [\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ and $[\text{NiCl}_4]^{2-}$
132. Find total number of cations which form low spin and inner orbital octahedral complex on adding excess NH_3 solution
 $\text{Cu}^{2+}(\text{aq}), \text{Co}^{2+}(\text{aq}), \text{Ir}^{3+}(\text{aq}), \text{Cr}^{3+}(\text{aq}), \text{Fe}^{3+}(\text{aq}), \text{Mn}^{2+}(\text{aq}), \text{Ti}^{3+}(\text{aq}), \text{Cd}^{2+}(\text{aq}), \text{Ni}^{2+}(\text{aq})$
133. Find the number of compounds in which $d_{x^2-y^2}$ orbital will take part in hybridization.
 (1) $[\text{PtBrCl}(\text{NH}_3)(\text{H}_2\text{O})]$ (2) SF_4 (3) $[\text{Cu}(\text{NH}_3)_4]^{2+}$
 (4) XeO_3F_2 (5) $[\text{RhCl}(\text{PPh}_3)_3]$ (6) $[\text{Co}(\text{en})_3]^{3+}$
 (7) $\text{Fe}(\text{CO})_5$ (8) $[\text{Cu}(\text{NO}_2)_5]^{3-}$ (9) XeF_4 (10) XeO_6^{4-}
134. Among following ligands, how many of them can get as π -acid ligand(s) with σ -donating ability and accepting back bond ability into its π^* M.O.
 $\text{CO}, \text{C}_5\text{H}_5^-, \text{B}_3\text{N}_3\text{H}_6, \text{PMe}_3, \text{cyclopropyl}, \text{C}_2\text{H}_4, \text{NO}^+, \text{F}^-, \text{O}_2^{2-}$
135. Find the value of expression $[x + y]$ for commercial extraction of following metals.
 Ag, Zn, Hg, Cu, Fe, Mg, Pb, Al, Au
 X : Number of metals commercially extracted by pyrometallurgy
 Y : Number of metals during their extraction chemical leaching is used
136. Find the value of expression $(x \div y)$ for complexes
 (A) $\text{Co}_2(\text{CO})_x$ (B) $\text{Fe}(\text{CO})_2(\text{NO})_y$
137. Find the total number of diamagnetic complex.
 $\text{K}_3[\text{Fe}(\text{CN})_6], [\text{Fe}(\text{EDTA})]^{2-}, \text{Na}_3[\text{Co}(\text{Ox})_3], [\text{Ni}(\text{H}_2\text{O})_6]\text{Cl}_2, \text{K}_2[\text{Pt}(\text{CN})_4], [\text{Zn}(\text{NO}_2)_4](\text{NO}_3)_2, \text{K}_2[\text{NiCl}_4],$
 $[\text{Pt}(\text{NH}_3)_4]\text{Cl}_2, [\text{Ni}(\text{CO})_4]$

138. Consider the following ligands and find the value of term (x + y)
 SCN^- , NO_2^- , gly, acac, bcac, $\text{S}_2\text{O}_3^{2-}$, OH^- , EDTA^{4-} , dmg
 X = Number of ligands which form five membered ring with metal cation in respective complex compound.
 Y = Number of ligands which form six membered ring with metal cation in respective complex compound.
139. Among the following sulphides, find the number of metal sulphides which are not soluble in yellow ammonium sulphide (YAS).
 CuS , Bi_2S_3 , As_2S_3 , SnS_2 , CdS , SnS , Sb_2S_3 , PbS , Sb_2S_5
140. Among the following metals, the number of elements that release H_2 gas on reaction with non oxidizing acid and strong alkali solution.
 Be , Al , Cu , Mg , Ca , Zn , Sn , Pb , Fe
141. Predict the number of compounds that can give positive chromyl chloride test on reaction with solid $\text{K}_2\text{Cr}_2\text{O}_7$ and few drops of conc. H_2SO_4 .
 CaCl_2 , NH_4Cl , FeCl_3 , ZnCl_2 , AgCl , PbCl_2 , MnCl_2 , CuCl_2 , HgCl_2 .
142. Determine the sum of molar coefficient of SO_2 and KMnO_4 , when they react with each other in acidic medium.
143. Find the total number of ligands among the following species which can exhibit both ambidentate and flexidentate character separately.
 SCN^- , CN^- , NO_2^- , $\text{S}_2\text{O}_3^{2-}$, SO_3^{2-} , SO_4^{2-} , NO_3^- , NOS^- , OCN^-
144. Among the following, the number of cation(s) that give precipitate with NaCl solution is
 $\text{Ag}^+(\text{aq})$, $\text{Cu}^{2+}(\text{aq})$, $\text{Fe}^{3+}(\text{aq})$, $\text{Mg}^{2+}(\text{aq})$, $\text{K}^+(\text{aq})$, $\text{Hg}_2^{2+}(\text{aq})$, $\text{Pb}^{2+}(\text{aq})$, $\text{Al}^{3+}(\text{aq})$, $\text{Ca}^{2+}(\text{aq})$
145. Among the following, the number of compounds/ions having a linear structure is :
 C_3O_2 , N_3^- , $(\text{CN})_2$, I_3^+ , IF_2^- , $(\text{SCN})_2$, O_3 , N_2O , NH_2^- .
146. Consider the following sequence of reaction

$$\text{KMnO}_4 + \text{KOH}(\text{conc}) \longrightarrow \underset{\substack{\text{Green} \\ \text{solution}}}{\text{X}} + \text{H}_2\text{O} + \text{O}_2 ; \quad \text{X} + \text{H}_2\text{O} \longrightarrow \underset{\substack{\text{light} \\ \text{Pink}}}{\text{Y}} + \underset{\substack{\text{Dark} \\ \text{Brown}}}{\text{Z}} \downarrow$$

 Then calculate value of $|\text{P} + \text{Q} - \text{R}|$ where, P, Q and R are oxidation states of central atoms of X, Y and Z respectively.
147. Total number of π -acid (π -acceptor) ligands among given species is/are
 NH_2^- , CO , NO , PR_3 , C_2H_4 , C_6H_6 , CN^- , H_2O , C_5H_5^- , H^-
148. Calculate value of $|x - y|$ where x and y are total number of maximum equivalent 'Cr – O' bonds in CrO_5 and $\text{Cr}_2\text{O}_7^{2-}$ ion respectively.
149. Total number of acidic radical(s) which does/do not give white precipitate with AgNO_3 solution
 $\text{SO}_3^{2-}(\text{aq})$, $\text{S}_2\text{O}_3^{2-}(\text{aq})$, $\text{C}_2\text{O}_4^{2-}(\text{aq})$, $\text{I}^-(\text{aq})$, $\text{PO}_4^{3-}(\text{aq})$, $\text{CrO}_4^{2-}(\text{aq})$, $\text{NO}_2^-(\text{aq})$, $\text{S}^{2-}(\text{aq})$, $\text{MnO}_4^-(\text{aq})$
150. Calculate value of $|\text{P} - \text{Q}|$, where
 P : Sum of oxidation states of all sulphur atoms in tetrathionic acid
 Q : Sum of oxidation states of all sulphur atoms in thiosulphuric acid
151. In electrolytic refining of impure copper containing impurities of Ag, Hg, Al, Zn, Au, Fe, Pt, Sn, Mg. Then find out total number of metals present in anode mud in their elemental form.
152. How many of the following species would have C–O bond length longer than that of free CO molecules.
 $[\text{V}(\text{CO})_6]^-$, $[\text{Ti}(\text{CO})_6]^{2-}$, $\text{BF}_3(\text{CO})$, $[\text{Sc}(\text{CO})_6]^{3+}$, $\text{Ni}(\text{CO})_4$, $[\text{Fe}(\text{CN})_5(\text{CO})]^{3-}$, CO_2 , CO_3^{2-} , $[\text{Cr}(\text{CO})_6]$

Answer Key - Advanced Chemistry Assignment-3B

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1	2	3	4	5	6	7	8	9	10	11	12
D	D	B	C	A	D	C	A	C	C	D	D
13	14	15	16	17	18	19	20	21	22	23	24
C	C	B	C	D	D	A	A	C	D	B	B
25	26	27	28	29	30	31	32	33	34	35	36
D	A	D	A	A	B	C	B	D	D	A	C
37	38	39	40	41	42	43	44	45	46	47	48
B	B	D	B	ABD	AB	ABD	ABC	ABC	ABC	AC	ABD
49	50	51	52	53	54	55	56	57	58	59	60
ABC	ABCD	ABCD	ABCD	ACD	BCD	AC	ABCD	ABCD	ABC	BCD	ABD
61	62	63	64	65	66	67	68	69	70	71	72
ABD	BD	BCD	BCD	ABD	CD	ABC	ABCD	ACD	AB	ACD	ABCD
73	74	75	76	77	78	79	80	81			
BC	ABCD	ABCD	BCD	CD	BCD	ACD	ABD	A-PQR ; B-Q S ; C-PQ ; D-PR			
82				83				84			
A-PQRST ; B-PQST ; C-PQRT ; D-PQRT				A-PQRST ; B-PQ ; C-QRT ; D-QRST				A-QRS ; B -PQ ; C-PQRT ; D-ST			
85				86				87			
A-PST ; B-RS ; C-PT ; D-PQS				A-QRS ; B-QPT ; C-ST ; D-PQ				A -PQRT ; B -PRST ; C-PQRST ; D-RS			
88				89				90			
A-PQRT ; B-PST ; C-PQRST ; D-PQRST				A-PT ; B-RT ; C-PQT ; D-QRST				A-PQRST ; B-PQ RT ; C-QR ; D-QRST			
91				92				93			
A-QRS ; B-PQ RS ; C-PQT ; D-PQR				A-PQST ; B-R ; C-PQRST ; D-PQRT				A-PRST ; B-PQ RS ; C-PRS ; D-PQRST			
94				95				96			
A-PRS ; B-PQ ; C-QRS ; D-QRST				A-PRT ; B-PQ ; C-PQRST ; D-PQT				A-PRS ; B-PQRST ; C-QT ; D-PQRST			
97				98	99	100	101	102	103	104	105
A-PQR ; B-PQT ; C-PQRS ; D-PQRT				6	3	6	8	2	4	3	6
106	107	108	109	110	111	112	113	114	115	116	117
7	3	4	7	4	3	4	2	3	0	4	6
118	119	120	121	122	123	124	125	126	127	128	129
5	5	5	5	1	6	5	2	2	6	5	7
130	131	132	133	134	135	136	137	138	139	140	141
2	7	2	7	3	8	4	6	6	4	5	6
142	143	144	145	146	147	148	149	150	151	152	
7	2	3	5	9	7	2	4	6	4	7	